CIBC COLLECTIONS HACKATHON · FINALS

Inside the AI engine of Collections Copilot

How the models turn 1M customers, 760K notes and call audio into a cited next-best action, and what goes in and out at every layer.

CLASSIFIER	MATCHER	LLM	SPEECH
Hardship LR	**Entity resolution**	**Gemini, schema-bound**	**Whisper, on-device**

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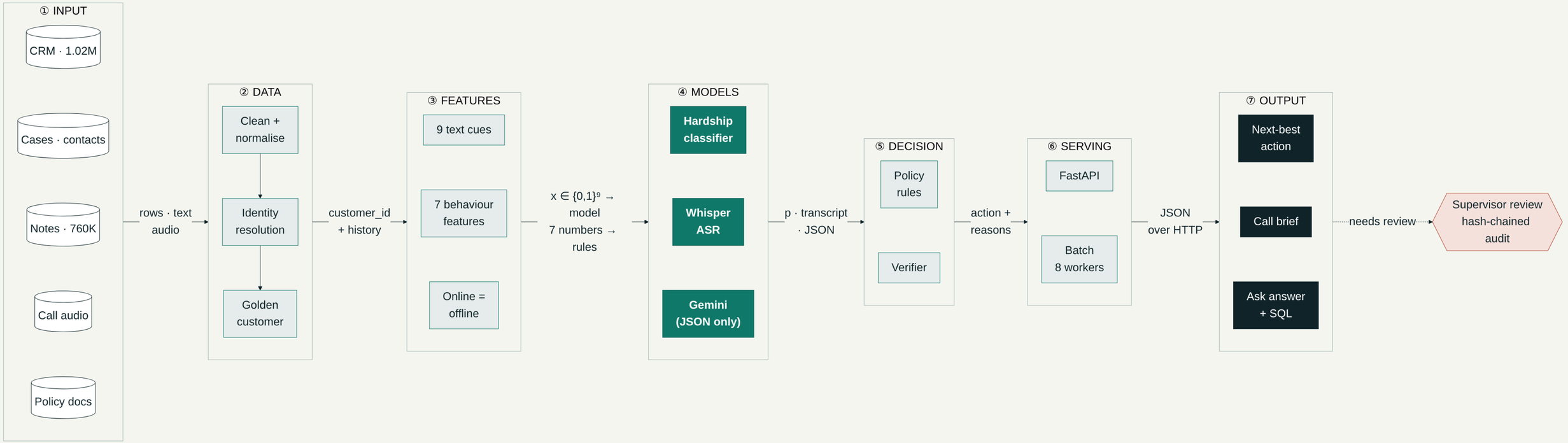
THE ML PROBLEM

Four learning tasks, each a typed input → output

QUESTION	INPUT	MODEL	OUTPUT
Who is this person?	3.83M source records: name, DOB, phone, postal	**Blocking + scored matcher with a margin test**	customer_id + confidence, or a review flag
Is the customer in hardship?	Free-text agent notes, EN + FR (760K)	**9 text cues → logistic regression**	p(hardship) + the quoted sentence
What is the supervisor asking?	A plain-English question	**Gemini → plan JSON (never SQL)**	Reviewed-SQL answer, SQL shown
What happened on the call?	Call audio and the agent's note	**Whisper ASR + Gemini extractor**	Verified fields: promise, dispute, hardship

ARCHITECTURE

Seven layers; every arrow is a typed payload

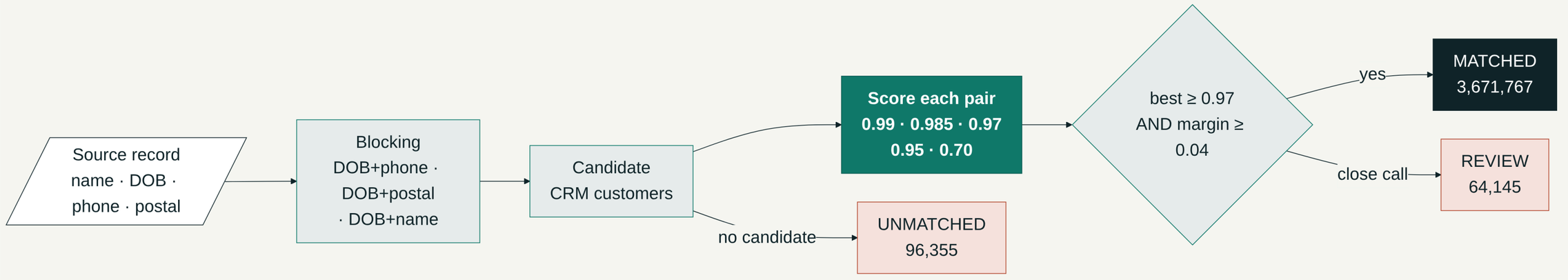


Input · Learned or neural model · Deterministic code · Output · Human review

Diagram source: Mermaid, in the repo

MODEL 1 · ENTITY RESOLUTION

Linking 3.83M records to one golden customer



Matched 95.8%

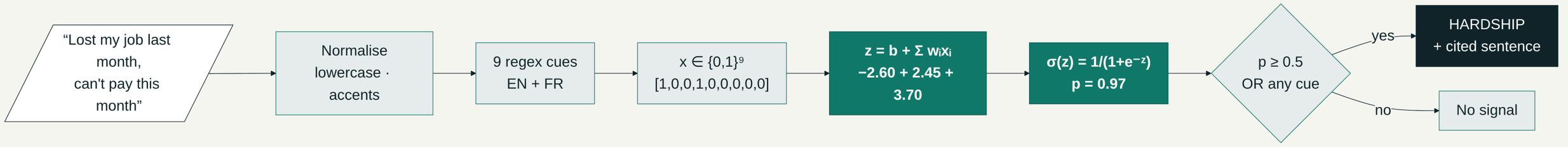
1,014,631 golden customers from 1,020,000 CRM rows

Review 1.7% · **Unmatched 2.5%**

scripts/build_data.py · fuzzy string similarity is shown to reviewers only

MODEL 2 · HARDSHIP CLASSIFIER

From one sentence to a probability



INPUT VECTOR x · THE ONLY THING THE MODEL SEES

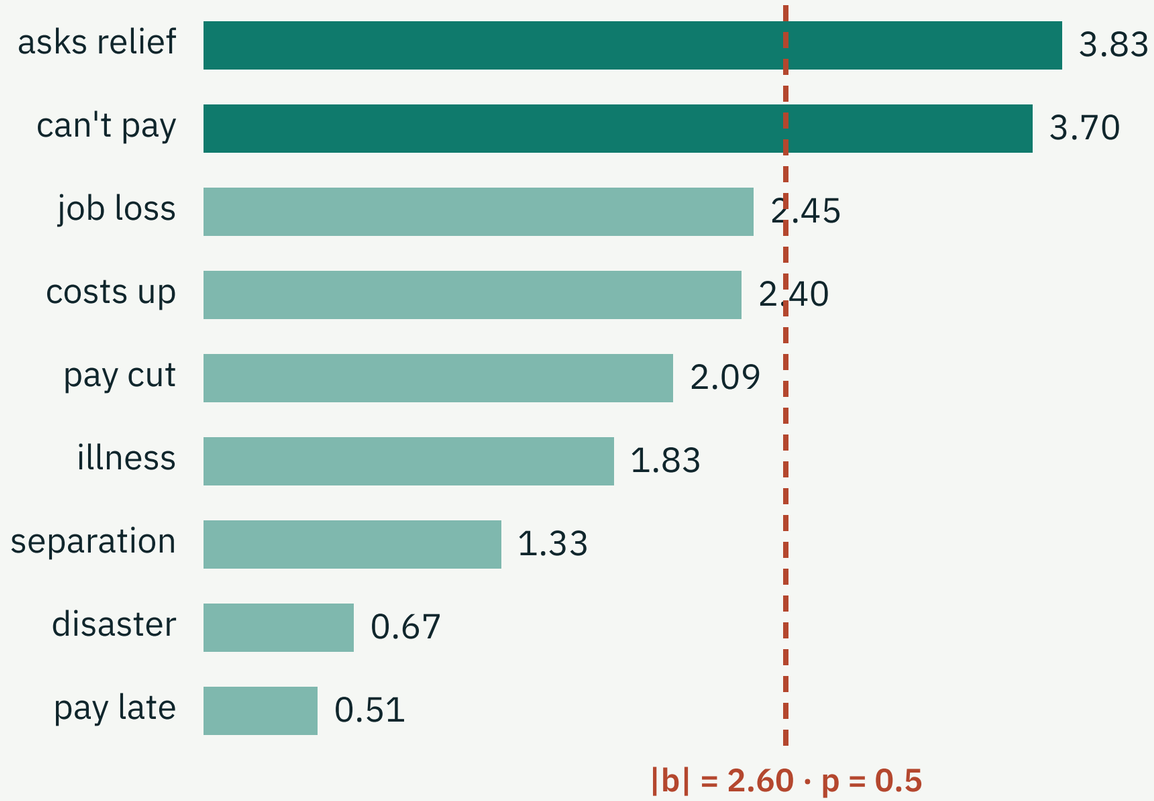
1	0	0	1	0	0	0	0	0
job loss	pay cut	pay late	can't pay	asks relief	costs up	illness	separation	disaster

Never in x: raw text, age, gender, language, vulnerability tags

Output: p = 0.97 → HARDSHIP, citing “lost my job last month”

WHAT THE MODEL LEARNED

Only two cues cross the decision line alone



job loss + can't pay

`−2.60 + 2.45 + 3.70 = 3.55`

p = 0.97 · hardship

pay late, alone

`−2.60 + 0.51 = −2.09`

p = 0.11 · cue rule still flags it

no cue

`−2.60`

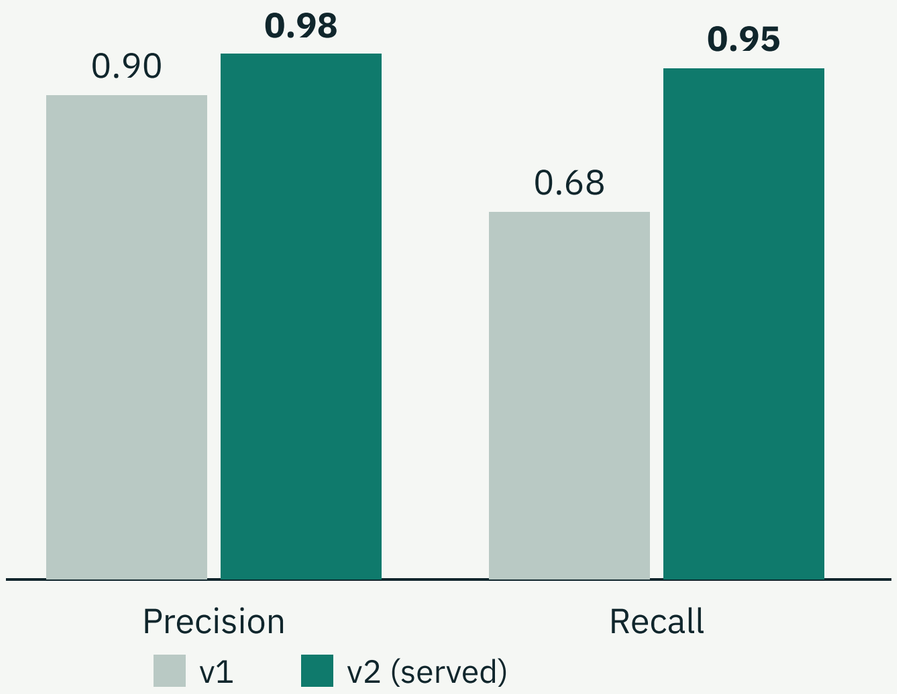
p = 0.07 · no signal

Class-balanced logistic regression, C = 1.0 · served rule: p ≥ 0.5 OR any cue

EVALUATION

Measured on customers the model never saw

Cross-validation, 5 folds × 5 seeds



Held-out confusion matrix

	pred. no	pred. yes
true no	88	0
true yes	2	36

38/38

held-out hardship notes caught by the served rule, model OR cue

500 labelled notes, 146 positive. Split by **customer**, so no person is in both train and test.

scripts/evaluate_hardship_cv.py · StratifiedGroupKFold · 374 train / 126 held-out, seed 1590

SMALL MODEL VS LLM · SAME 126 HELD-OUT NOTES

For hardship flags, nine weights beat Gemini

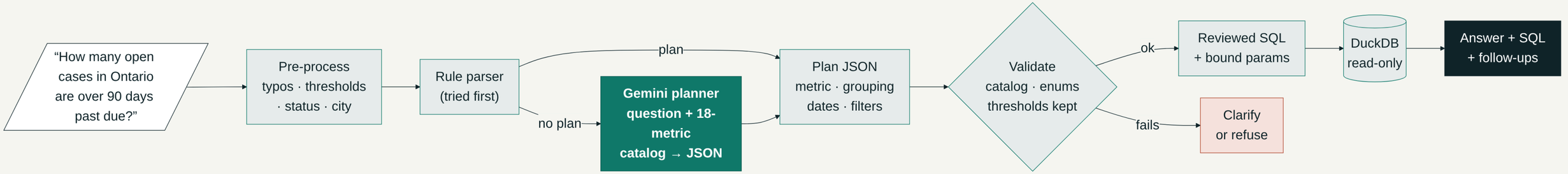
MEASURE	OUR CLASSIFIER	GEMINI 3.5 FLASH-LITE
Hardship notes caught	38/38	38/38
False alarms	0	4
Time per note (median)	< 1 ms	1.4 s
All 760,594 notes	< 13 min	12.3 days

Gemini's false alarms: 3 notes carried the “cust vulnerable – financial hardship” tag, a vulnerability marker that policy keeps separate from hardship, and 1 was a consumer proposal. So the LLM goes where language is open-ended, not here.

scripts/compare_llm_classifier.py → data/llm-comparison.json · Gemini stable over 40 reruns

LLM 1 · ASK, A CONSTRAINED PLANNER

The LLM fills a form; it never writes SQL



① PLAN JSON (SCHEMA-BOUND)

```
{"metric_id": "open_cases",
 "grouping": "none",
 "start_date": "2026-09-28",
 "end_date": "2026-09-29",
 "filters": {"province": "ON",
  "dpd": {"op": ">", "value": 90}}}
```

② REVIEWED SQL, BOUND PARAMS

```
SELECT count(*) open_cases
FROM curated.collections_cases c
WHERE c.outcome IS NULL
 AND c.current_dpd > ?
 AND c.province_code = ?
params: [90, "ON"]
```

③ OUTPUT

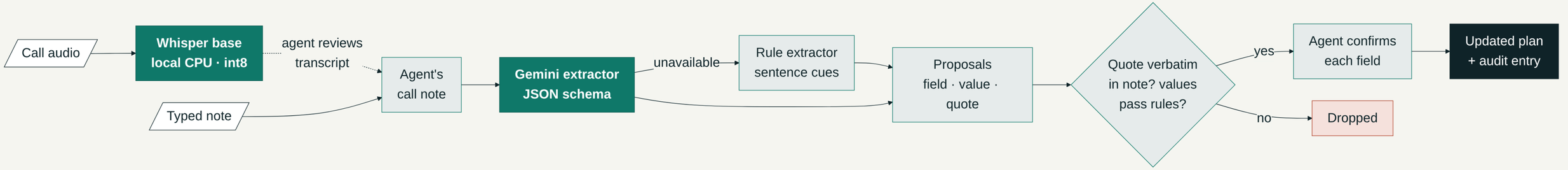
7,681

open Ontario cases over 90 days past due · 81 ms

app/insights.py · app/metric_plans.py · 18-metric catalog · thresholds dropped by a plan → rejected

LLM 2 · CALL-NOTE EXTRACTION

Every extracted field must quote the note



INPUT NOTE

“Customer said she was **laid off last month**. She can **pay $200 on Friday**. Asked us to **call back after 6pm**.”

GEMINI PROPOSALS → VERIFIER

✓ promise · $200 · 2026-10-02 · quote found

✓ hardship · job_loss · quote found

✗ promise · $500 · amount not in quote

✗ “Customer lost her job.” · not verbatim

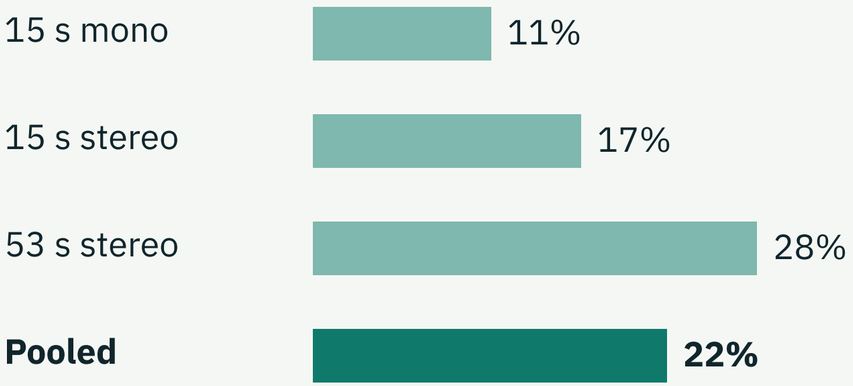
app/call_session.py · verify_model_proposals · nothing is saved until the agent confirms

MODEL 3 · SPEECH RECOGNITION

Call audio is transcribed on the laptop

WAV · 2,000 calls → SHA-256 checked → **faster-whisper base · CPU int8** → cache: audio + model hash → segments + language

Word error rate, 3 exploratory samples



2.7 s to transcribe a 53 s call on CPU, no audio leaves the machine

3/3 hardship cue vectors from the transcript equal those from the reference text

app/voice.py · data/voice-evaluation.json · synthetic TTS samples, not a held-out test · review-only

DECISION LAYER · ML SIGNAL + POLICY RULES

Models raise signals; cited rules decide

INPUTS

hardship: recorded / possible / none

flags: insolvency, deceased, cease, legal

contacts in last 7 days, time zone

promises, payday, programmes

RULES, IN ORDER · EACH STEP CITES ITS POLICY

1 · Holds stop everything	POL-COLL-001 §7.3
2 · Representative or cease request	POL-COLL-001 §5.4
3 · Vulnerable-customer approach	POL-COLL-004 §4.5
4 · Hardship → offer programmes first	POL-COLL-004 §4.2
5 · Contact window: 07–21 h, cap 3 per 7 days	§5.1 · §5.2
6 · Payment ask timed to payday	PRC-COLL-010 §3.1

```
→ {"next_action": "Offer one of these hardship programmes first", "policy": ["POL-COLL-004 §4.2"], "evidence": ["note sentence"]}
```

app/action_plan.py · “Rules only; no model output changes a step”

INFERENCE AT SCALE · MLOPS

Whole-portfolio inference in 21 seconds

93,320
action plans in 21.1 s, 8 workers

4,430/s
batch throughput

0.17 s
live call update via the API

0
online/offline feature mismatches

MODEL LIFECYCLE · EVERY STEP IS A SCRIPT WITH A SAVED REPORT

Build data
raw → curated → gold

→

Train
versioned model file

→

Cross-validate
grouped, 5 × 5

→

Parity check
1,000 × 18 columns

→

Serve + batch
598 tests pass

scripts/build_data.py → train_model.py → evaluate_hardship_cv.py → check_feature_parity.py → batch_plans.py

RESPONSIBLE AI

Audited for bias, bounded against hallucination

Model recall by group, held-out



Gender is audited only, never a model input

- **No protected data** in the feature vector
- **JSON schema** on every LLM call
- **Verbatim quotes** checked against the source
- **Notes are data:** embedded instructions ignored
- **Counts under 10** shown as “fewer than 10”
- **Human confirms**; hash-chained audit log

data/model-report.json · restricted_fairness_audit · small groups: read as a warning sign, not a verdict

IN ONE LINE PER LAYER

Small models where decisions repeat; LLMs where language is open; rules where policy speaks

38/38
held-out hardship notes caught

0
lines of SQL written by the LLM

21 s
to plan 93,320 customers

Questions · we can open any layer live